

CAPE LISBURNE, AK, USA

WMO: 701040

Lat: 68.875N

Lon: 166.113W

Elev: 16

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26631

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-23.5	-35.0	0.8	-27.0	-30.8	1.0	-23.3	53.9	20.3	46.5	20.0	7.3	110	1.040

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	61.5	54.2	58.1	52.5	55.5	51.3	55.3	59.4	53.3	57.2	51.5	55.2	14.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.2	60.2	57.3	50.9	55.2	55.3	49.1	51.7	53.6	23.4	59.6	22.1	57.4	21.2	55.3	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
41.1	35.7	31.7	-27.6	68.0	8.7	4.2	-33.9	71.0	-39.0	73.5	-43.9	75.9	-50.3	78.9		
			-27.8	58.2	7.8	2.2	-33.4	59.8	-38.0	61.1	-42.3	62.4	-48.0	64.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	-1.7	-0.4	-1.7	12.3	29.5	39.1	48.3	47.5	40.2	29.0	15.9	4.5
	DBStd	21.38	14.82	16.38	14.11	11.18	7.78	6.51	5.39	5.05	6.19	7.36	10.94	13.30
	HDD50	10318	1604	1412	1603	1132	636	334	97	111	300	653	1024	1412
	HDD65	15704	2069	1832	2068	1582	1101	776	519	544	745	1117	1474	1877
	CDD50	89	0	0	0	0	0	9	43	32	5	0	0	0
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH74	2	0	0	0	0	0	0	0	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	13.0	13.0	13.6	12.6	11.8	10.8	9.7	12.4	13.0	14.3	15.5	15.0	13.8
	PrecAvg	18.2	1.2	1.2	0.7	0.7	0.7	1.1	2.2	3.1	2.1	2.3	1.6	1.1
Precipitation	PrecMax	24.7	3.0	3.0	2.2	2.1	1.7	2.0	3.9	5.9	3.5	3.9	4.6	2.3
	PrecMin	12.4	0.1	0.1	0.1	0.1	0.0	0.5	0.7	0.7	0.4	0.7	0.3	0.1
	PrecStd	3.1	0.7	0.8	0.6	0.6	0.4	0.4	0.7	1.3	0.7	0.8	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	39.2	35.7	42.3	51.0	64.7	65.2	70.3	58.2	49.4	43.0	40.9
		MCWB	34.2	34.8	30.3	36.1	44.3	53.2	56.0	64.4	51.2	41.8	39.4	33.4
	2%	DB	33.7	33.9	32.1	37.7	45.2	57.4	61.6	60.3	53.7	44.4	37.1	34.2
		MCWB	31.4	31.6	28.6	34.5	40.6	50.7	54.4	55.5	49.5	39.8	34.8	30.9
	5%	DB	29.1	31.0	28.7	33.8	42.0	52.6	58.9	56.5	51.5	40.8	34.0	30.2
		MCWB	26.8	28.9	25.7	31.4	38.8	47.4	52.8	53.1	48.1	38.2	32.2	28.0
	10%	DB	23.2	26.3	22.9	28.4	38.8	48.9	56.5	54.4	49.4	38.0	30.4	24.8
		MCWB	21.3	24.2	20.5	26.5	36.4	45.0	51.6	51.4	46.5	35.7	28.5	22.8
	0.4%	WB	34.6	34.7	30.9	37.0	44.5	54.6	57.5	65.4	52.5	43.0	39.2	35.1
		MCDB	37.0	38.0	32.8	40.8	49.9	62.4	62.0	69.7	55.7	47.2	42.5	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.3	31.7	28.0	34.4	41.4	51.1	55.2	56.3	50.3	40.5	35.0	31.2
		MCDB	33.5	33.9	30.6	37.8	44.5	57.7	59.9	59.5	53.4	43.6	36.6	34.7
	5%	WB	26.9	28.9	24.6	31.5	38.7	47.9	53.4	53.5	48.6	38.2	32.3	28.0
		MCDB	28.8	30.8	27.6	33.7	41.4	52.4	57.9	56.3	51.3	40.7	34.0	30.4
	10%	WB	21.0	23.9	18.9	26.7	36.6	45.1	51.9	51.5	46.4	35.9	28.6	22.9
		MCDB	23.0	26.1	21.2	28.5	38.7	49.0	56.0	54.1	49.1	38.3	30.3	24.9
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.4	10.0	10.1	7.3	8.6	8.4	6.3	5.5	5.8	7.5	8.7
		MCDBR	15.7	14.2	12.1	13.1	12.6	15.1	12.8	10.0	8.3	7.9	8.2	12.2
		MCWBR	14.3	12.7	11.2	11.0	9.0	9.7	8.5	7.0	5.9	6.4	6.9	10.5
	5% WB	MCDBR	15.4	14.3	13.4	13.2	11.8	14.4	11.7	9.3	7.9	7.6	8.0	12.5
		MCWBR	14.0	13.1	11.9	11.4	8.7	9.7	8.3	7.0	5.8	6.4	6.9	10.9
Clear-Sky Solar Irradiance	taub		0.162	0.221	0.252	0.323	0.356	0.343	0.359	0.367	0.310	0.253	0.358	N/A
	taud		1.903	2.047	2.188	2.010	2.061	2.399	2.452	2.389	2.498	2.212	2.848	N/A
	Ebn at Noon		24	176	249	253	259	271	262	243	232	162	39	0
	Edh at Noon		2	14	23	39	42	31	29	27	19	12	3	0
All-Sky Solar Radiation	RadAvg		8	143	675	1366	1702	1842	1459	962	547	164	20	0
	RadStd		1	17	50	80	202	145	146	120	55	24	2	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page